

How to use the UARTBOOT terminal program

MMSP™ 2

Application Note

MDMP2520FAN200504

Ver. 1.0 / September 26,2005

MP2520F

*How to use the UARTBOOT terminal program Ver 2.0
for MMSP 2 Application Processor*

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1. UARTBOOT

1.1. What is the UARTBOOT

UARTBOOT is one of the booting modes of MP2520F. DNW v2.0 is a terminal program for UARTBOOT & USB downloader. By using of it, user can download 'NANDBOOT image file' and 'EBOOT(or Magicloader) image file' from PC to NAND flash. There is no need to use JFLASH or Multi-ICE.

1.2. The precondition of the UARTBOOT

- Supported port : UART port 0 of MP2520F
- Supported speed : 38400 bps only
- Download data size : 16K or 32K (it is possible to select the download size by setting the RstCfgNFBSize pin)

1.3. The Functions of the UARTBOOT terminal program

- Function as a terminal of UART Boot loader
- Function as a USB downloader (can download Bootloader file from PC to NAND flash through USB 1.1 device port)

1.4. The Preparations for using the UARTBOOT terminal program

- **Caution) Below four files must be located in the same folder**
 - DNW Terminal program (MES-DNW_V2.0)
 - UARTBOOT.BIN File
 - NANDBOOT File
 - Bootloader File (ex Magicloader or EBOOT file)
- USB cable
- UART cable
- USB driver for **DNW v2.0**

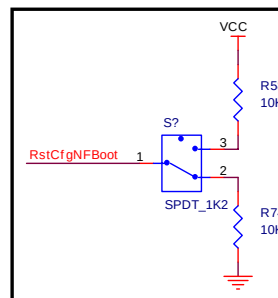
1.5. Initial Setting for using the UARTBOOT

Related Configuration Pin	Description
RstCfgNFBoot	Boot select 0:NANDBOOT disable 1:NANDBOOT enable
RstCfgNFBSize	NANDBOOT size 0: 512byte, 1: 512byte UARTBOOT size 0 : 16KB, 1: 32KB
RstCfgBootDown	UART boot enable 0:UARTBOOT disable 1:UARTBOOT enable

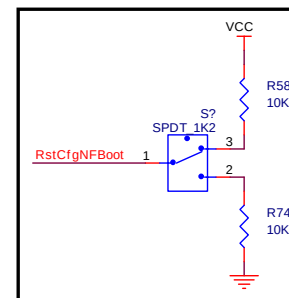
For NANDBOOT case, **RstCfgNFBSize** and **RstCfgBootDown** pin is meaningless so user can use NANDBOOT by setting **RstCfgNFBoot** pin to 1(High).

1.6. Hardware Modification point for using the UARTBOOT on DTK411, DTK2nd, and DTK3rd

- DTK411
 - R75 remove
 - R76 remove
 - R60 10K Install
 - R59 10K Install
 - Add SW (Fig.1)



UARTboot



NANDboot

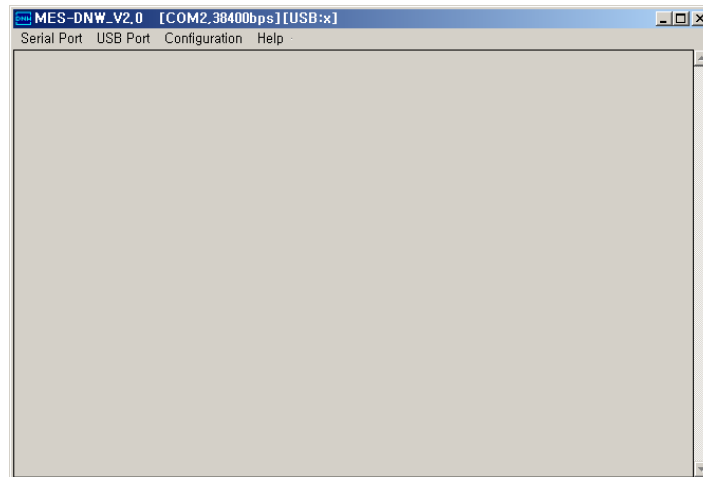
<Fig.1>

- DTK2, DTK3

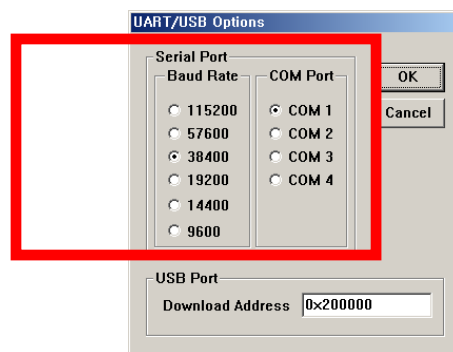
UARTBOOT	NANDBOOT
S4 → L	S4 → H
S5 → H	S5 → H
S18 → H	S18 → H

2. The Usage of the UARTBOOT terminal program

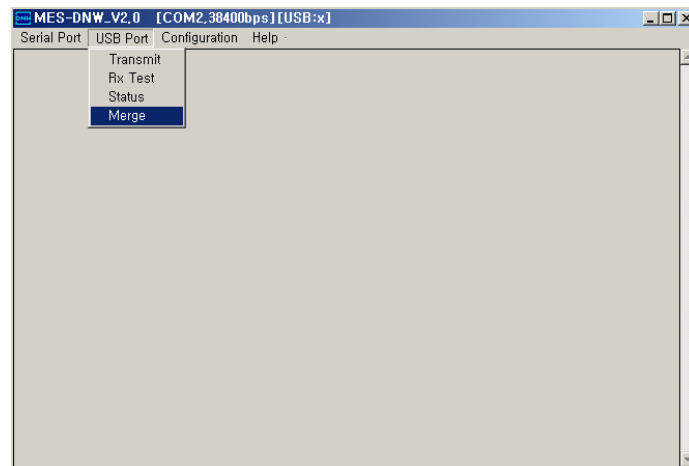
- 1 Connect the Serial Cable(UART Cable) between PC and KIT
- 2 Execute the DNW v2.0 Terminal



- 3 Select **Configuration** → **Option** and set the Baud Rate & COM port of PC (Baud Rate : 38400bps)



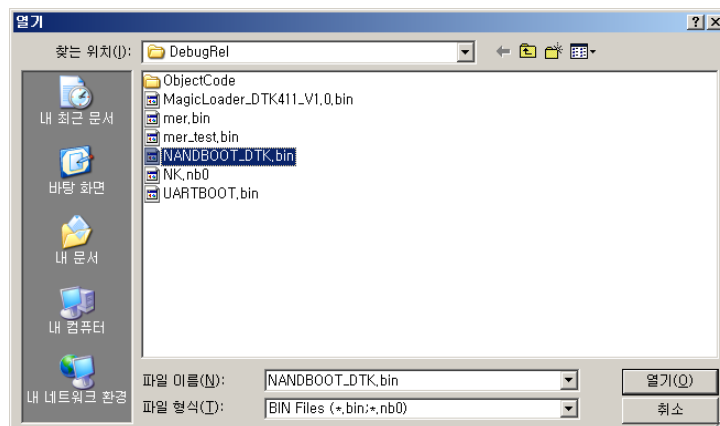
- 4 Select **USB Port** → **Merge (NANDBOOT and Bootloader)**



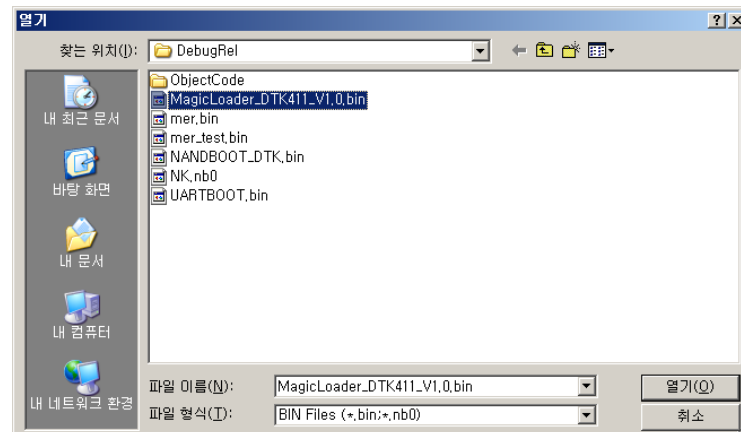
5 Select the **NANDBOOT & Bootloader File**.

- Selecting order : **NANDBOOT File** → **Bootloader File**

5.1 select the NANDBOOT file and open it. (ex: 'NANDBOOT_DTK.bin')



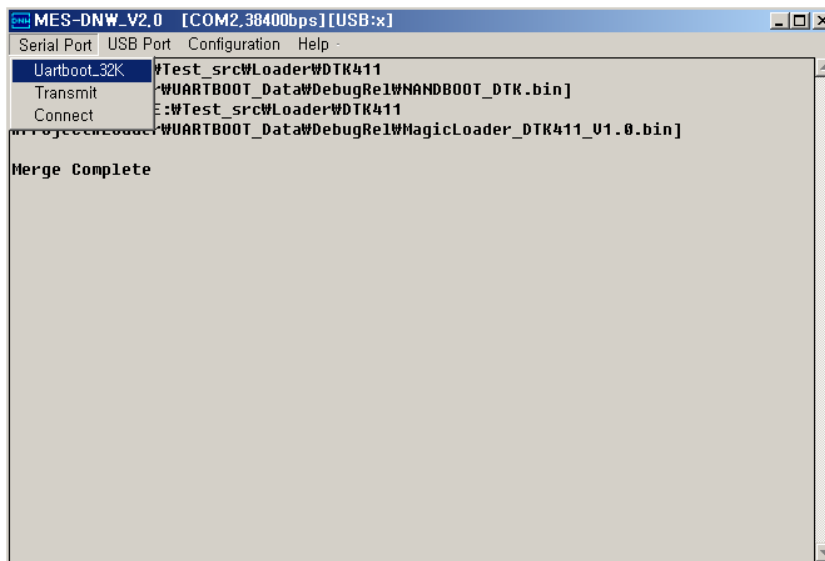
5.2 select the Bootloader file and open it. (ex: 'MagicLoader_DTK411_V1.0.bin')



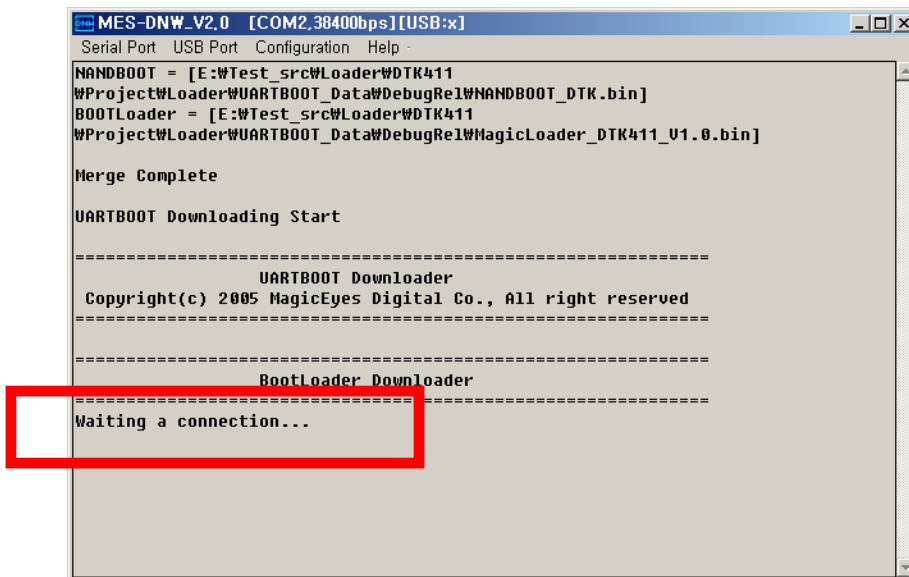
How to use the UARTBOOT terminal program (DNW v2.0)

- 6 **'mer.bin' file will be created automatically.**
- 7 After change the boot setting of KIT for UART Boot, Turn on the KIT Power

8 Select **Serial Port** → **Uartboot_32K** (ALT+S)



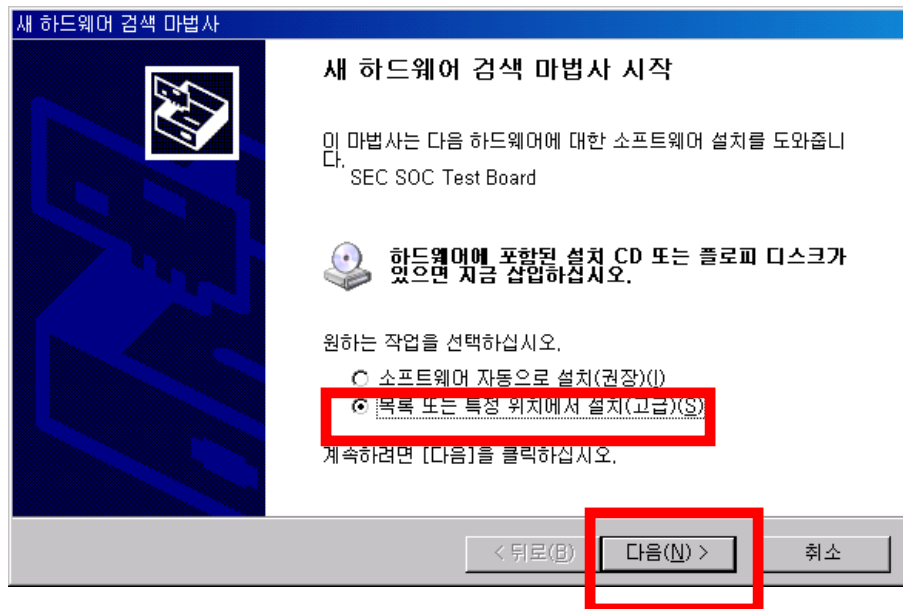
9 **Download Bootloader thru USB** : User can download Bootloader file through USB1.1 device port



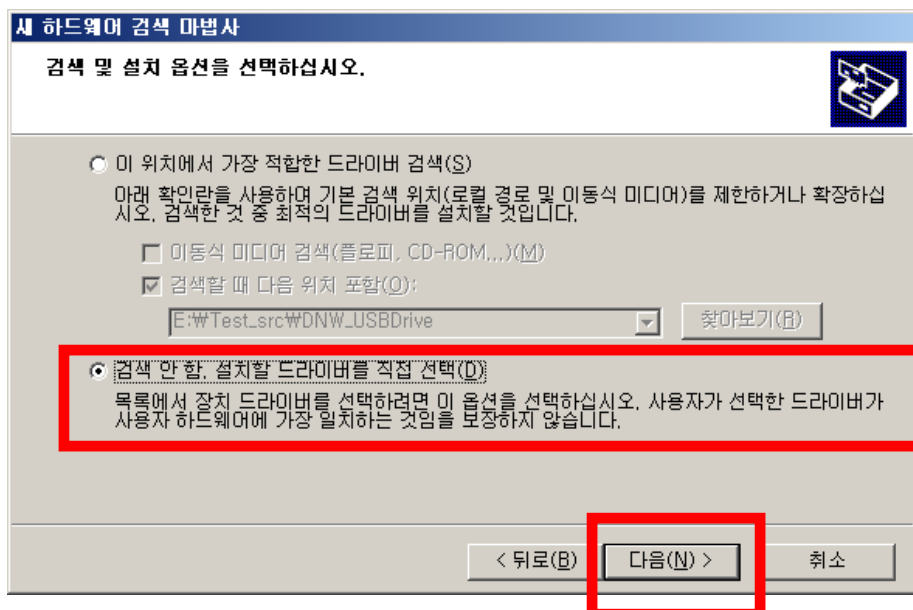
- 9.1 'Waiting a connection...' message will be printed in DNW terminal. Connect the USB cable between PC and KIT.

How to use the UARTBOOT terminal program (DNW v2.0)

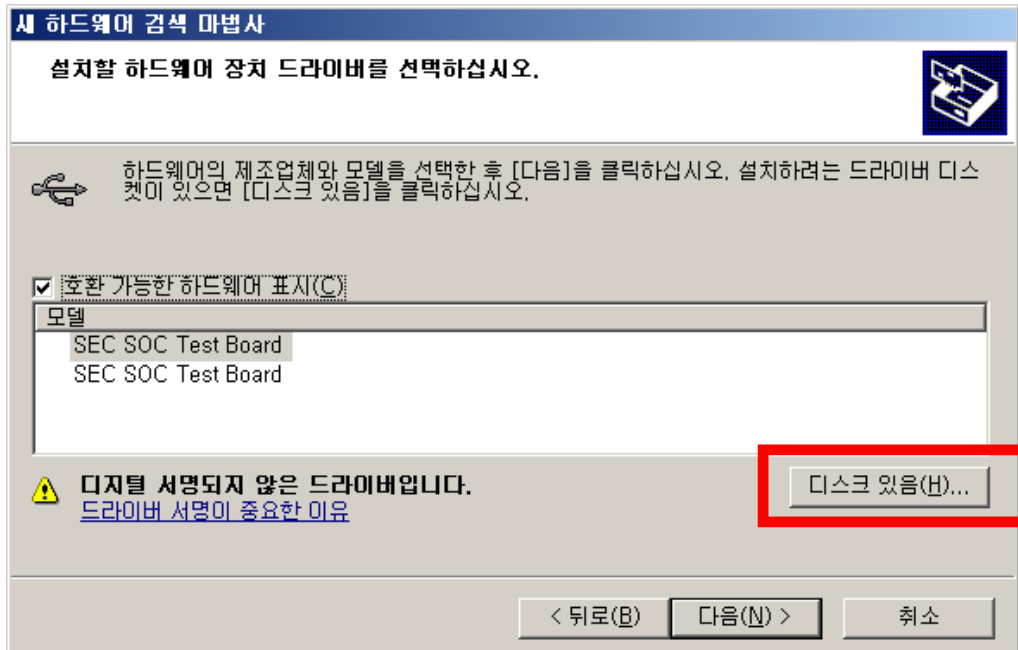
- 9.2 PC will detect new hardware and 'New Hardware Searching Wizard' window will be popped up. There will be a two options. Do not select 'auto installation' option, select 'installation from list' option and click 'Next' button.



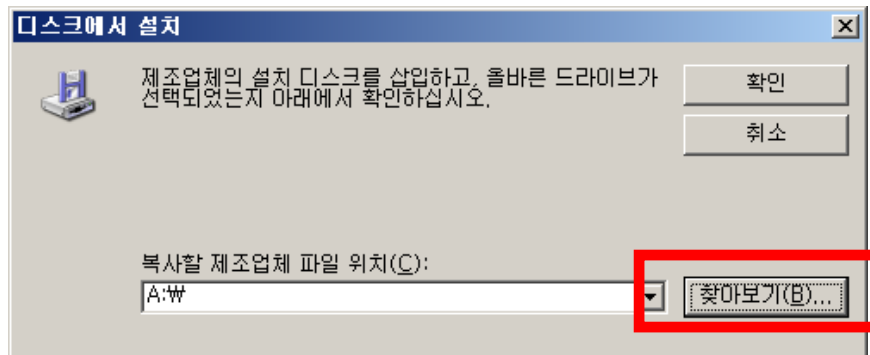
- 9.3 Select 'Not search. Select driver directly' option and click 'Next' button



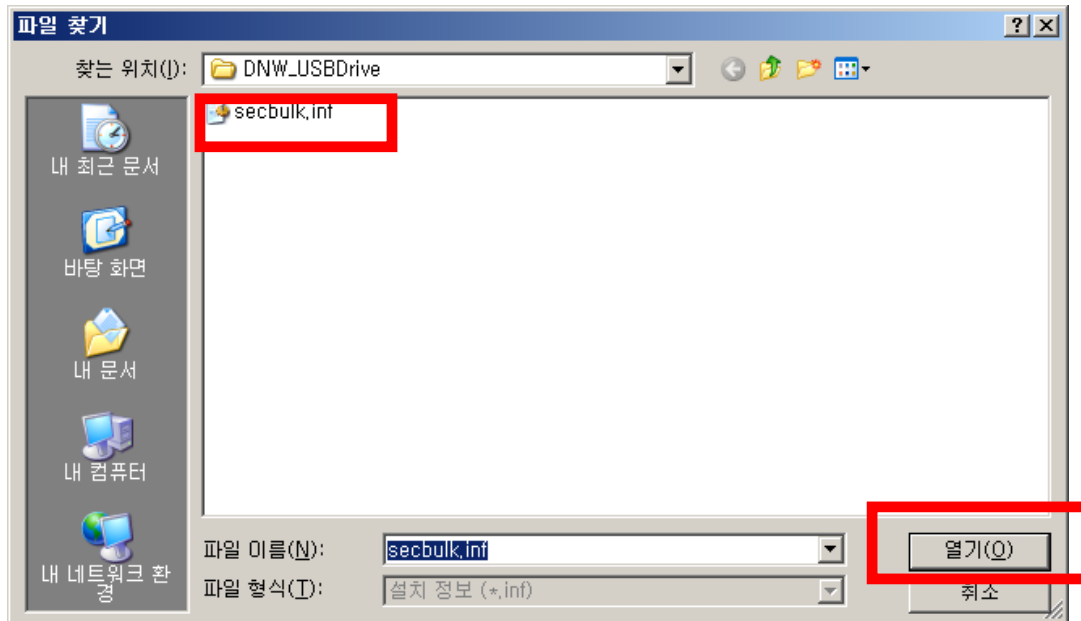
9.4 Click 'Disk' button



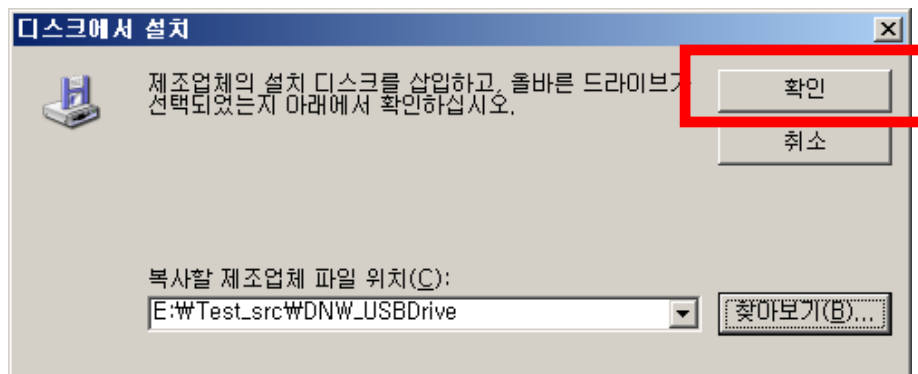
9.5 Click 'Search' button.



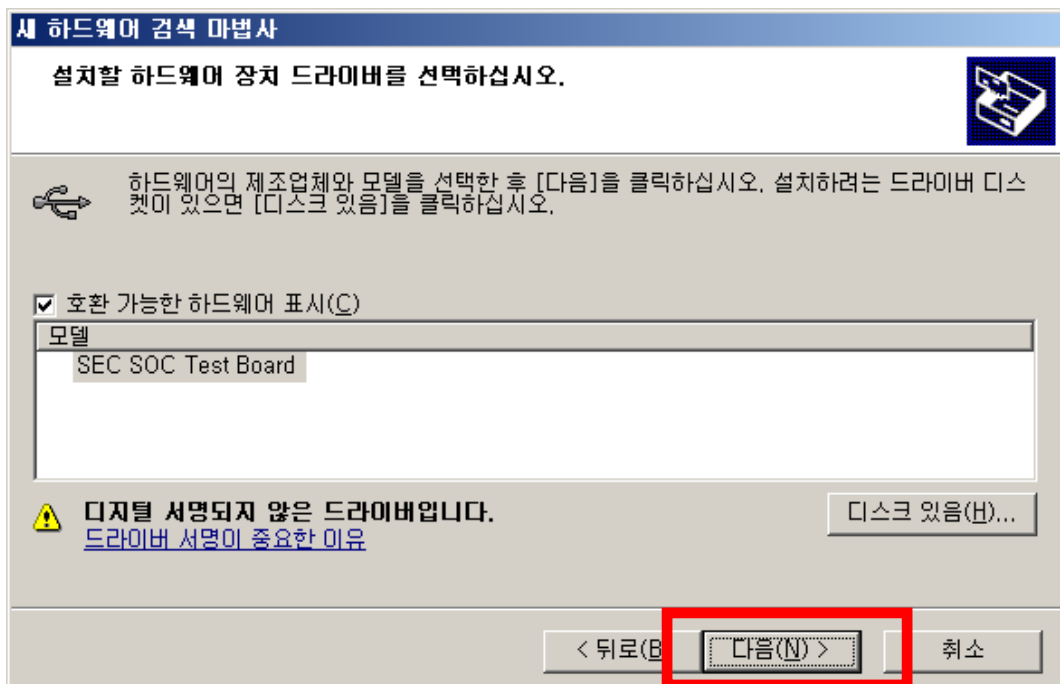
9.6 Select 'secbulk.inf' file and open it.



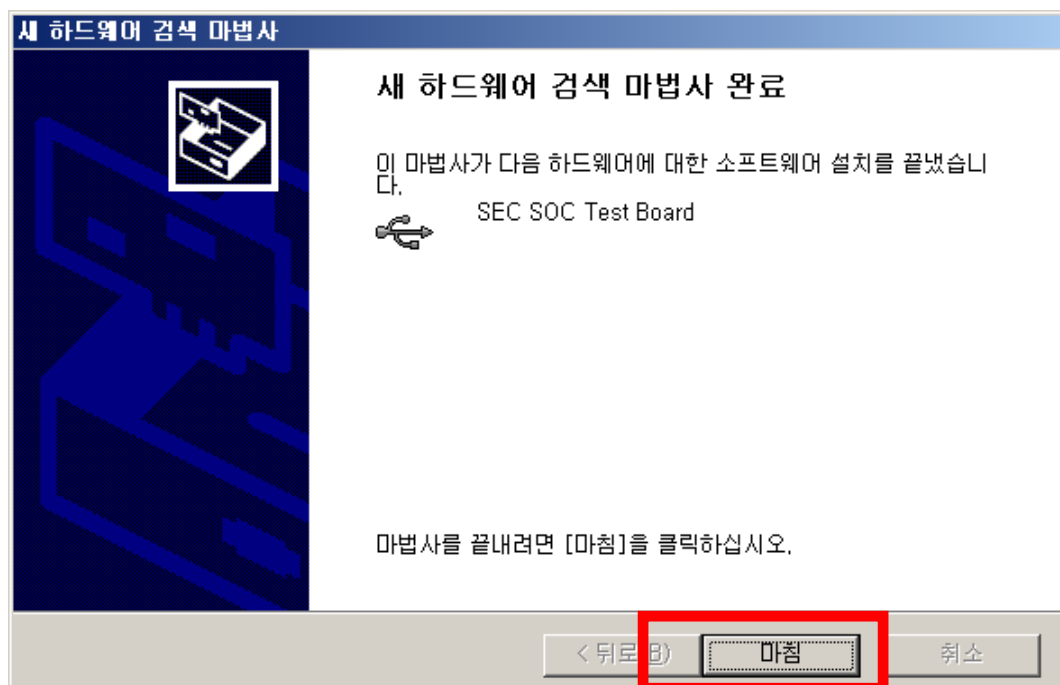
9.7 Click 'Confirm'(or OK) button



9.8 Click 'Next' button



9.9 Click 'Finish' button



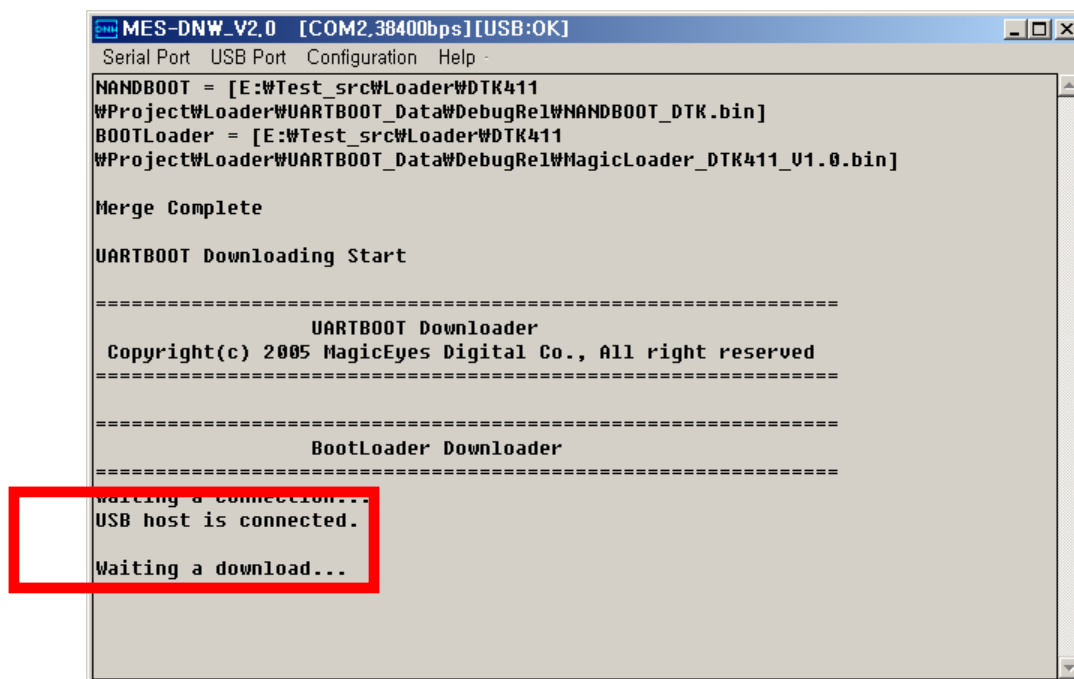
How to use the UARTBOOT terminal program (DNW v2.0)

9.10 Click 'Finish' button

9.11 If the driver installation is finished successfully, you can see the below message :

'USB host is connected.'

'Waiting a download...'



```
MES-DNW_V2.0 [COM2,38400bps][USB:OK]
Serial Port  USB Port  Configuration  Help

NANDBOOT = [E:\Test_src\Loader\WDTK411
WProject\Loader\UARTBOOT_Data\DebugRel\NANDBOOT_DTK.bin]
BOOTLoader = [E:\Test_src\Loader\WDTK411
WProject\Loader\UARTBOOT_Data\DebugRel\MagicLoader_DTK411_U1.0.bin]

Merge Complete

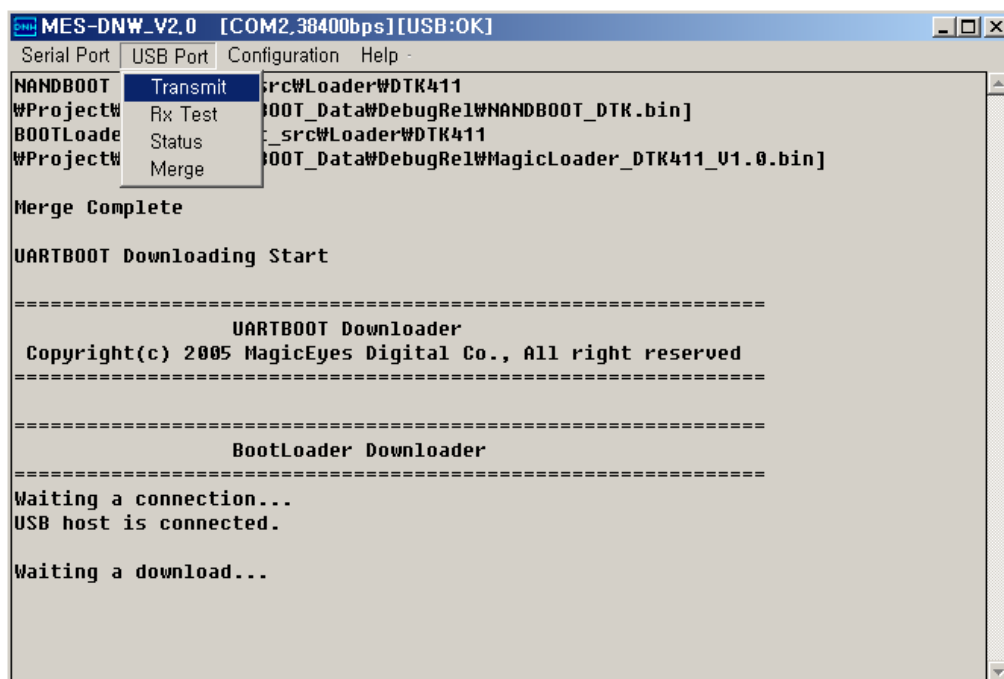
UARTBOOT Downloading Start

=====
                UARTBOOT Downloader
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=====

                BootLoader Downloader
=====

Waiting a connection...
USB host is connected.
Waiting a download...
```

9.12 Click **USB Port** → **Transmit** and select the 'mer.bin' file to download.



```
MES-DNW_V2.0 [COM2,38400bps][USB:OK]
Serial Port  USB Port  Configuration  Help

NANDBOOT      Transmit  src\Loader\WDTK411
WProject\    Rx Test   BOOT_Data\DebugRel\NANDBOOT_DTK.bin]
BOOTLoader    Status   _src\Loader\WDTK411
WProject\    Merge    BOOT_Data\DebugRel\MagicLoader_DTK411_U1.0.bin]

Merge Complete

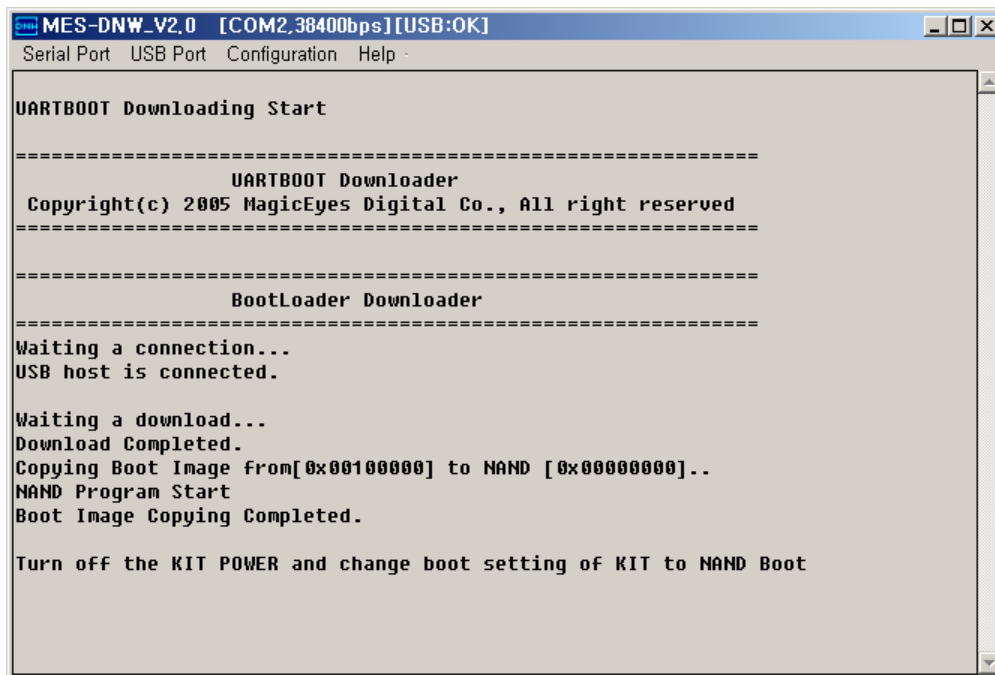
UARTBOOT Downloading Start

=====
                UARTBOOT Downloader
Copyright(c) 2005 MagicEyes Digital Co., All right reserved
=====

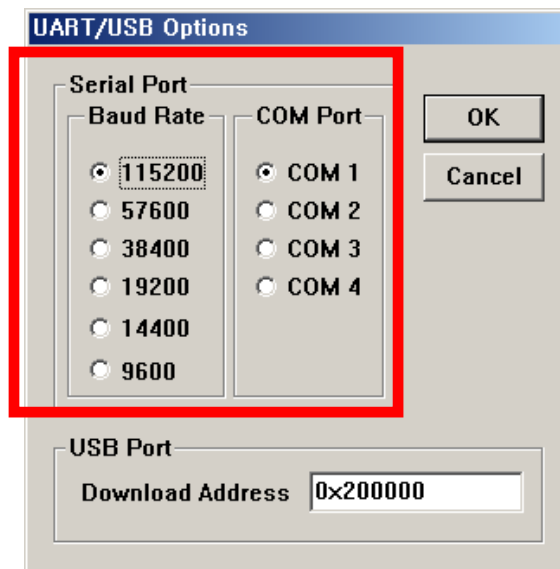
                BootLoader Downloader
=====

Waiting a connection...
USB host is connected.
Waiting a download...
```

10 Download Complete

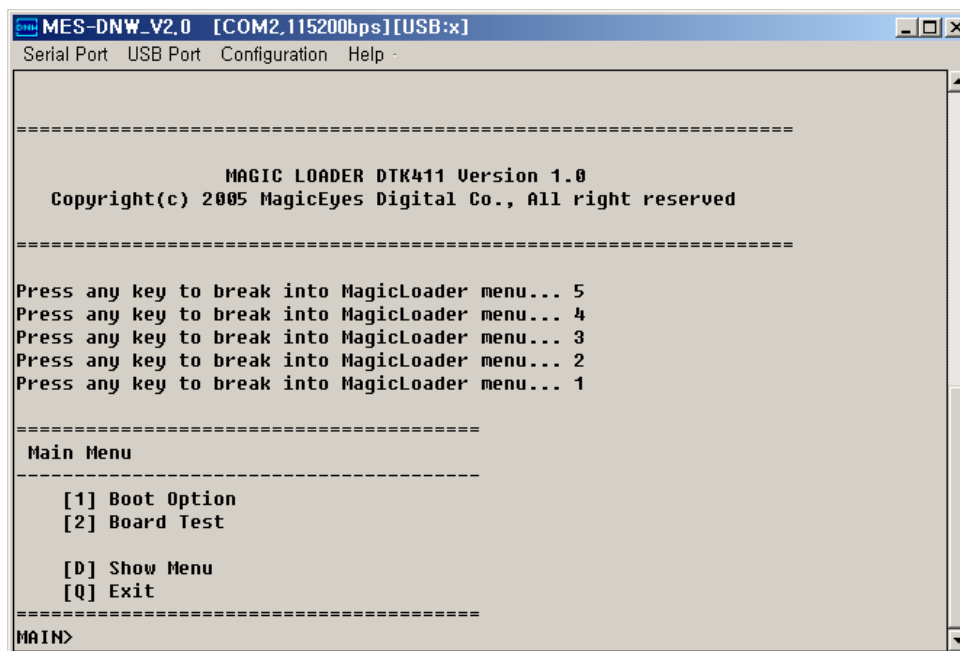


- 11 Turn off the KIT Power and change boot setting of KIT to NAND boot
- 12 Select **Configuration** → **Option** and set the COM port of PC and bootloader Baud rate (ex : 115200)



How to use the UARTBOOT terminal program (DNW v2.0)

- 13 Turn on the KIT Power. Bootloader Start



The screenshot shows a terminal window titled "MES-DNW_V2.0 [COM2,115200bps][USB:x]". The window has a menu bar with "Serial Port", "USB Port", "Configuration", and "Help". The main text area displays the following content:

```
=====
                        MAGIC LOADER DTK411 Version 1.0
                Copyright(c) 2005 MagicEyes Digital Co., All right reserved
=====

Press any key to break into MagicLoader menu... 5
Press any key to break into MagicLoader menu... 4
Press any key to break into MagicLoader menu... 3
Press any key to break into MagicLoader menu... 2
Press any key to break into MagicLoader menu... 1
=====

Main Menu
=====
    [1] Boot Option
    [2] Board Test

    [D] Show Menu
    [Q] Exit
=====
MAIN>
```